

MAP estimation and priors

Multiply the likelihood by what you believed before the data, then report the peak of the result.

THE QUESTION

With few observations the MLE overfits. How should prior knowledge move the estimate, and when does it stop mattering?

THE INTUITION

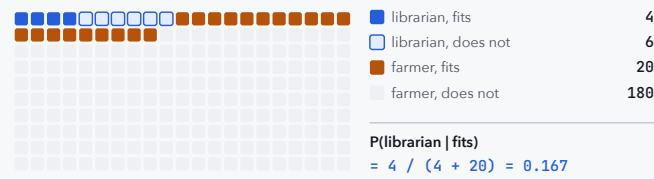
Bayes' rule weighs each θ by its prior, so in logs the log-prior adds to the log-likelihood. A Beta prior acts like extra tosses; a prior on weights acts like a penalty.

KEY TAKEAWAY

MAP maximises log-likelihood plus log-prior. A Gaussian prior gives ridge, a Laplace prior lasso. The prior's weight fades as N grows.

prior $p(\theta)$ → likelihood $p(D | \theta)$ → multiply → posterior $p(\theta | D)$ → mode: MAP → or mean, interval → = penalised loss

1 The base rate can outweigh a good description

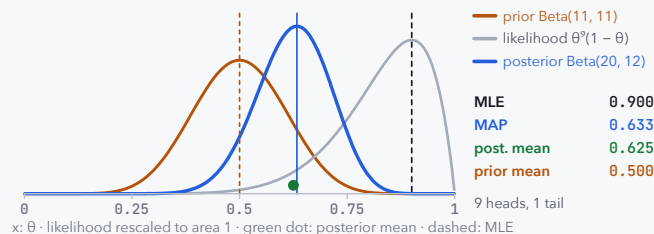


A shy, tidy person fits 40% of librarians and 10% of farmers, and there are 20 farmers per librarian:

$$P(L | d) = \frac{0.4 \cdot 1/21}{0.4 \cdot 1/21 + 0.1 \cdot 20/21} = \frac{0.0190}{0.0190 + 0.0952} = \frac{1}{6}$$

The likelihood ratio is 4 for librarian; the prior odds are 1 to 20, so farmer stays five times more likely.

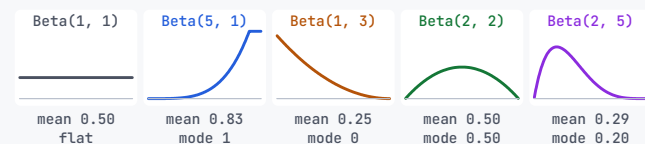
2 The posterior is prior times likelihood



$$p(\theta | D) = \frac{p(D | \theta) p(\theta)}{p(D)}, \quad p(D) = \int p(D | \theta) p(\theta) d\theta$$

$p(D)$ does not depend on θ , so $\theta_{\text{MAP}} = \arg \max_{\theta} p(D | \theta) p(\theta)$; the MLE drops $p(\theta)$. Nine heads in ten give $\theta_{\text{MLE}} = 0.9$; a Beta(11, 11) belief in a fair coin pulls the peak to 0.633. The 95% posterior interval [0.454, 0.781] contains 0.5.

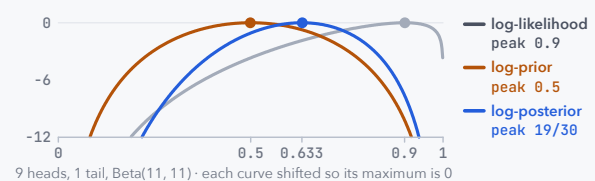
3 A Beta prior on [0, 1] has two shape counts



$$p(\theta | \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Mean $\alpha/(\alpha + \beta)$; mode $(\alpha - 1)/(\alpha + \beta - 2)$ when $\alpha, \beta > 1$. Beta(1, 1) is uniform. A larger $\alpha + \beta$ is a more confident prior at the same mean.

4 Zero slope of the log-posterior gives the MAP



With n_H heads and n_T tails, up to a constant,

$$\log p(\theta | D) = (n_H + \alpha - 1) \log \theta + (n_T + \beta - 1) \log(1 - \theta)$$
$$\frac{n_H + \alpha - 1}{\theta} = \frac{n_T + \beta - 1}{1 - \theta} \Rightarrow \theta_{\text{MAP}} = \frac{n_H + \alpha - 1}{N + \alpha + \beta - 2}$$

The mode counts $\alpha - 1$ extra heads and $\beta - 1$ extra tails: $(9 + 10)/(10 + 20) = 0.633$.

5 Beta in, Beta out: the conjugate update



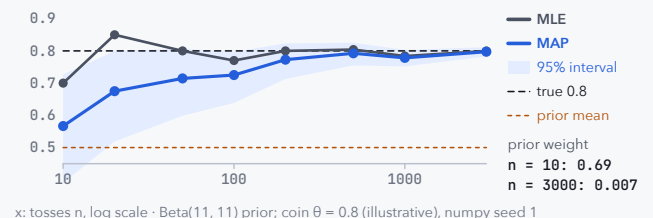
tosses HHTHHHHHTH (illustrative: $\theta = 0.7$, numpy seed 3), prior Beta(2, 2) curves after 0, 1, 3, 10 tosses; one batch update gives the same Beta(10, 4)

The posterior $\propto \theta^{n_H + \alpha - 1} (1 - \theta)^{n_T + \beta - 1}$ is again a Beta:

$$\text{Beta}(\alpha, \beta) \rightarrow \text{Beta}(\alpha + n_H, \beta + n_T)$$

Each toss adds 1 to one count: today's posterior is tomorrow's prior, and the order does not matter.

6 More data swamps the prior



The posterior mean mixes the two estimates:

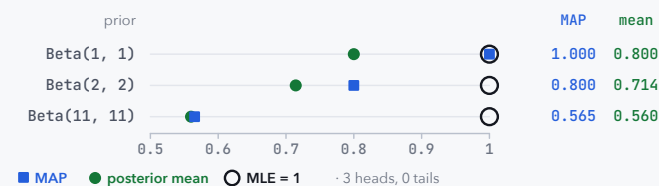
$$\frac{\alpha + n_H}{\alpha + \beta + N} = w \frac{\alpha}{\alpha + \beta} + (1 - w) \frac{n_H}{N}, \quad w = \frac{\alpha + \beta}{\alpha + \beta + N}$$

With $\alpha + \beta = 22$ the prior weight is 0.69 after 10 tosses and 0.007 after 3000; the MAP converges to the MLE and the interval narrows.

Priors on weights are penalties

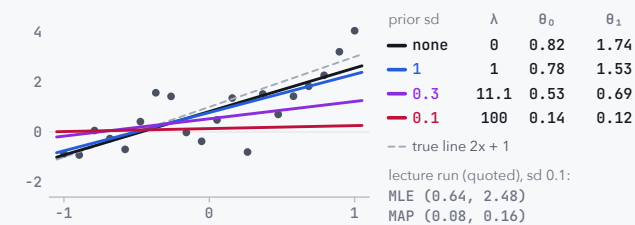
Mode against mean on a coin, then regression weights: a Gaussian prior is L2, a Laplace prior is L1.

7 Three heads in three tosses: MLE, MAP and mean



The MLE says tails "never" happen (the overconfident coin of the MLE sheet). The posterior is $\text{Beta}(\alpha + 3, \beta)$. With the flat $\text{Beta}(1, 1)$ the MAP is the MLE again, while the mean $(n_H + 1)/(N + 2) = 0.8$ is Laplace's rule of succession. The mean $(\alpha + n_H)/(\alpha + \beta + N)$ never reaches 0 or 1.

8 A Gaussian prior on the weights gives ridge



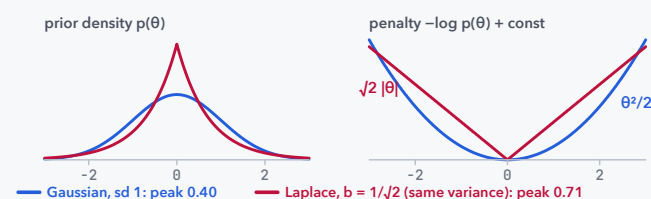
$y_i = \theta^T x_i + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, prior $\theta \sim \mathcal{N}(0, \sigma_0^2 I)$. The negative log-posterior is

$$\frac{1}{2\sigma^2} \|y - X\theta\|^2 + \frac{1}{2\sigma_0^2} \|\theta\|^2 + \text{const}$$

$$\Rightarrow \theta_{\text{MAP}} = (X^T X + \lambda I)^{-1} X^T y$$

with $\lambda = \sigma^2/\sigma_0^2$ (times σ^2 : $\frac{1}{2} \|y - X\theta\|^2 + \frac{\lambda}{2} \|\theta\|^2$). A narrow prior is a large penalty. The lecture's prior sd 0.1 ($\lambda = 100$) pulls both weights, intercept included, to about 0.1.

9 A Laplace prior gives the L1 penalty

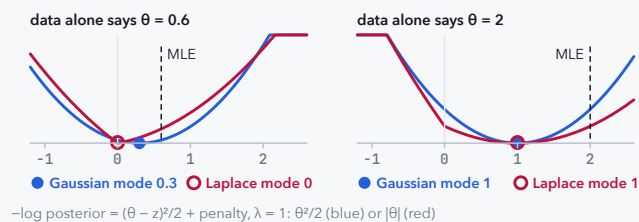


With $p(\theta_j) = \frac{1}{2b} \exp(-|\theta_j|/b)$ for each weight,

$$-\log p(\theta | D) = \frac{1}{2\sigma^2} \|y - X\theta\|^2 + \frac{1}{b} \sum_j |\theta_j| + \text{const}$$

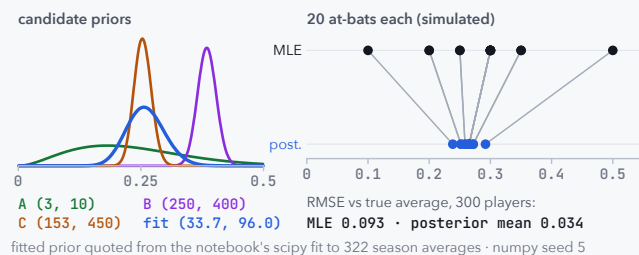
which is lasso: $\frac{1}{2} \|y - X\theta\|^2 + \lambda \|\theta\|_1$ with $\lambda = \sigma^2/b$. At equal variance the Laplace prior puts more mass near 0 and in the tails. In logistic regression both priors add the same penalties to the cross-entropy.

10 The kink holds small weights at exactly zero



One weight, data estimate z : the Gaussian mode $z/(1 + \lambda)$ is never 0 unless $z = 0$; the Laplace mode $\text{sign}(z) \max(|z| - \lambda, 0)$ sits at the corner for any $|z| \leq \lambda$. With 8 features, 3 of them real (30 points, seed 4, $\lambda = 8$), the Laplace MAP sets all 5 null weights to exactly 0 and the Gaussian MAP none. Zeros come from the mode: the Laplace posterior mean is never exactly sparse.

11 A fitted prior shrinks small samples



Fitted to 322 season batting averages, the prior is $\text{Beta}(33.7, 96.0)$, mean 0.260 (quoted). The posterior mean is $(h + 33.7)/(n + 129.7)$: 4 hits in 10 at-bats goes from 0.400 to 0.270; 170 in 631 only from 0.269 to 0.268. Candidate B (mean 0.385) sits where almost no player bats. At 20 at-bats shrinking cuts the error to about a third. A prior fitted to the same population is empirical Bayes.

12 Pause and think

Is MAP with a flat prior the MLE? Yes: $\log p(\theta)$ is constant.

Does the MAP change in log-odds? Yes: densities pick up a Jacobian. The log-odds mode of $\text{Beta}(20, 12)$ maps back to $20/32 = 0.625$, not 0.633. The MLE does not move.

What does a prior sd of 0.1 claim? Each weight lies within ± 0.2 with 95% probability. The slope is near 2, so the fit collapses.

How sure is the MAP? It is one point. The Bayesian linear regression sheet keeps the whole posterior.

MAP = argmax (log-lik. + log-prior) · Beta prior = extra tosses · Gaussian = L2, Laplace = L1 · more data, less prior

Source: lecture slides "Maximum a posteriori estimation" and the MAP, baseball and librarian notebooks, nipunbatra.github.io/pml-teaching. Numbers recomputed in NumPy with fixed seeds; the batting data and its fit are quoted. Fixed here: MAP minimises $\text{NLL} - \log p(\theta)$ (the slides write $\arg \min \text{NLL} + \log p(\theta)$; an older notebook coded that sign, so the slides' Laplace figure moves the slope away from 0, 2.55 to 2.71); the ridge term is $\|\theta\|^2/(2\sigma_0^2)$; the L1 term sums $|\theta_j|$; the Laplace density is in θ ; the MAP derivation's $\theta \sum (1 - y_i)$ term needs a minus.